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Course Code: DSA 0601
Course: Data Handling And Visualization

SET 27: PATIENT HEALTH RISK ANALYSIS

Q1. Scatterplot Matrix

Aim: examine pairwise relationships among Age, BMI, BP and Cholesterol.

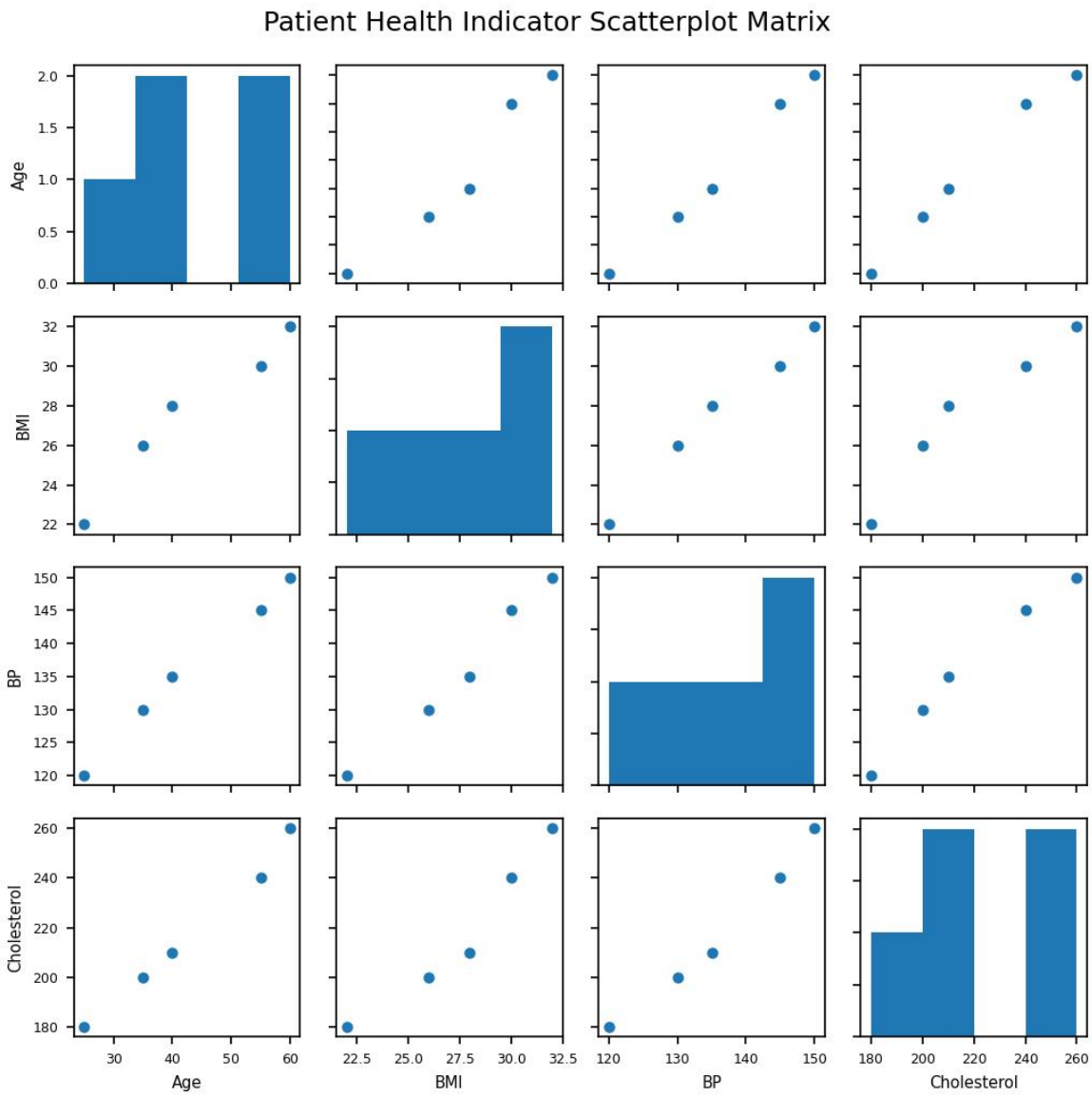
Algorithm:

1. Enter the five patient records.
2. Select the four numeric indicators.
3. Use pairs() and calculate correlations.

R Code:

```
patients <-
data.frame(Age=c(25,40,55,35,60),BMI=c(22,28,30,26,32),BP=c(120,135,145,130,150),Cholesterol=c(180,21
0,240,200,260))
health <- patients[,c("Age","BMI","BP","Cholesterol")]
pairs(health,main="Patient Health Indicator Scatterplot Matrix",pch=19,col="blue")
print(round(cor(health),2))
```

Output Visualization:



Result: The five records show generally positive pairwise relationships. The sample is too small for broad population conclusions.

Q2. Q-Q Plot and ECDF

Aim: assess the distribution of cholesterol and consider normality.

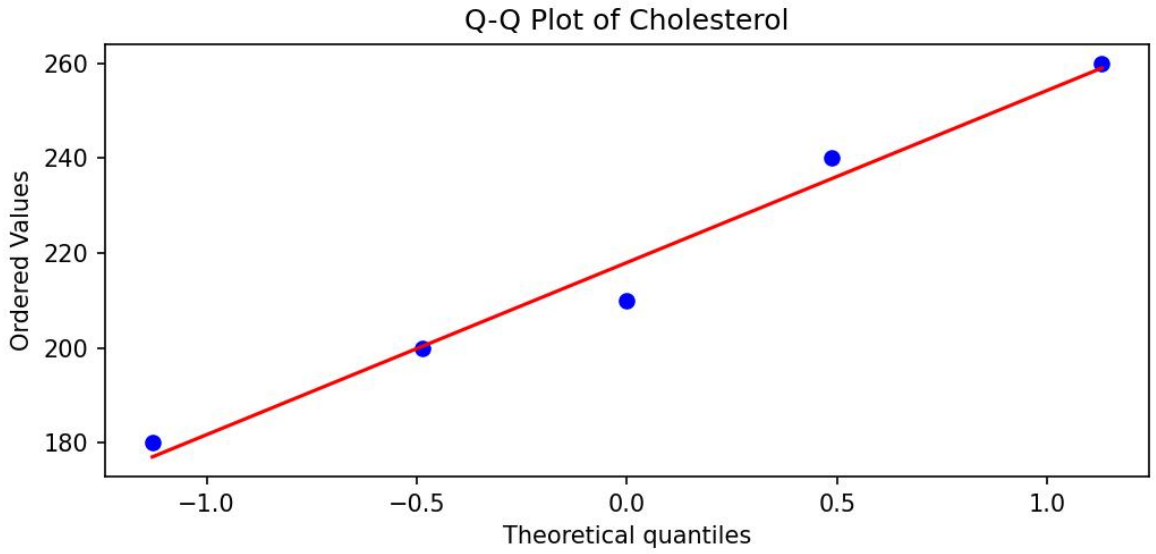
Algorithm:

1. Extract Cholesterol.
2. Draw a normal Q-Q plot with a reference line.
3. Draw the ECDF and inspect its shape.

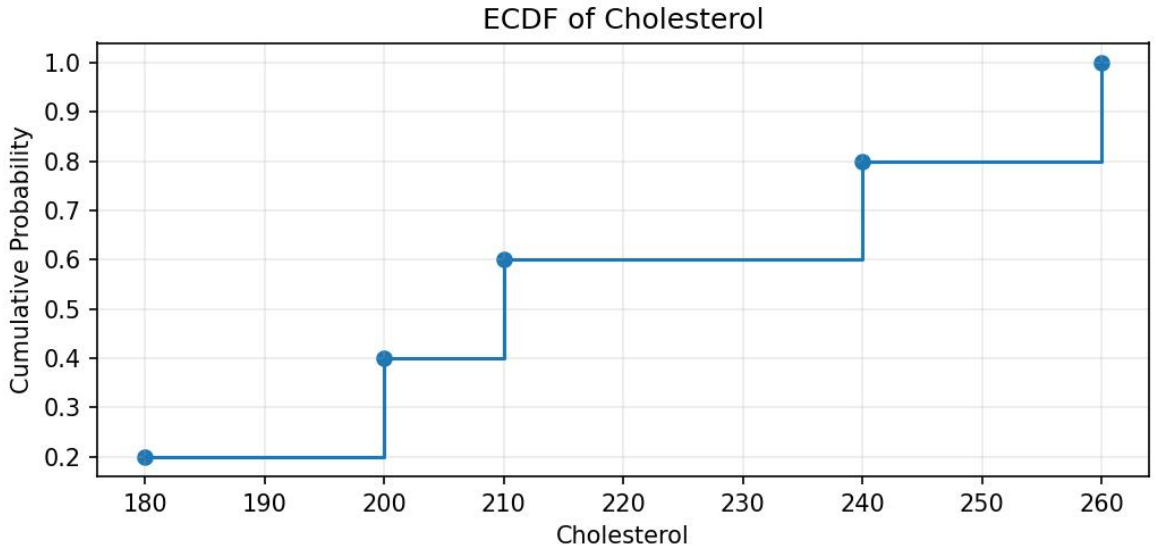
R Code:

```
chol <- patients$Cholesterol
qqnorm(chol,main="Q-Q Plot of Cholesterol",pch=19,col="blue"); qqline(chol,col="red",lwd=2)
plot(ecdf(chol),main="ECDF of Cholesterol",xlab="Cholesterol",ylab="Cumulative
Probability",verticals=TRUE,do.points=TRUE,col="darkgreen",lwd=2)
```

Q-Q Plot:



ECDF:



Result: The plots show theoretical quantiles and cumulative proportions. Five observations are insufficient to assess normality or skewness reliably.

Q3. Average Health Indicators

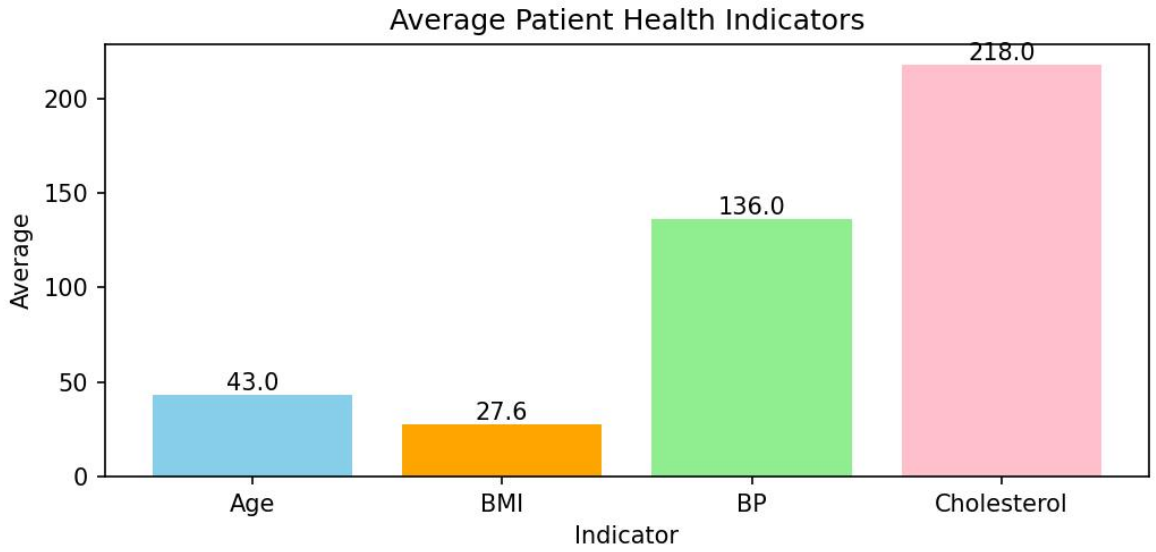
Aim: compare the average Age, BMI, BP and Cholesterol.

Algorithm:
1. Calculate the mean of each indicator.
2. Store the means in a named vector.
3. Draw a colored bar chart and label it.

R Code:

```
averages <-
c(Age=mean(patients$Age),BMI=mean(patients$BMI),BP=mean(patients$BP),Cholesterol=mean(patients$Choles
terol))
print(averages)
barplot(averages,col=c("skyblue","orange","lightgreen","pink"),main="Average Patient Health
Indicators",xlab="Indicator",ylab="Average Value",las=1)
```

Output Visualization:



Result: Mean Age = 43 years, BMI = 27.6, BP = 136 mmHg systolic, and Cholesterol = 218 (units not specified). These are descriptive averages, not diagnoses.

Q4. Tableau Interactive Heatmap

Aim: explore health indicators by age and highlight records needing further review.

Algorithm:
1. Import patient data into Tableau.
2. Build a heatmap by Patient_ID and health indicators.
3. Add Age filter and tooltips for all indicators.
4. Optionally flag BMI >= 30 OR BP >= 140 OR Cholesterol >= 240 for further review.
5. Combine the views in a dashboard.

Result: P5 has the highest BMI, BP and cholesterol; P3 also meets the illustrative review thresholds. This simple flag is not a validated medical risk score and must not replace professional judgment.